

YANCHI, China

WMO: 537230

Lat: 37.7925N

Lon: 107.3899E

Elev: 4449

StdP: 12.48

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -3.2	(c) 1.2	(d) -19.4	(e) 2.2	(f) 8.5	(g) -15.5	(h) 2.8	(i) 11.2	(j) 20.9	(k) 30.9	(l) 18.7	(m) 29.6	(n) 1.6	(o) 270	(p) 0.495

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 19.9	(c) 90.4	(d) 61.9	(e) 88.0	(f) 61.5	(g) 85.5	(h) 61.1	(i) 69.3	(j) 79.7	(k) 67.9	(l) 78.4	(m) 66.5	(n) 77.2	(o) 7.5	(p) 140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 66.6	(b) 115.7	(c) 73.6	(d) 64.7	(e) 108.2	(f) 72.8	(g) 62.9	(h) 101.5	(i) 71.9	(j) 36.5	(k) 79.8	(l) 35.1	(m) 78.4	(n) 33.9	(o) 77.5	(p) 76.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.8	16.2	13.9	-9.0	94.8	5.1	2.5	-12.6	96.7	-15.6	98.1	-18.5	99.5	-22.1	101.4
			-9.8	71.2	4.8	1.9	-13.3	72.6	-16.1	73.7	-18.8	74.8	-22.4	76.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.7	19.4	26.7	39.2	52.1	61.8	70.1	73.8	70.7	61.1	49.3	35.6	23.0		
	DBStd	20.02	7.92	8.83	9.05	8.37	6.58	5.06	4.90	5.33	6.23	7.03	7.93	7.71		
	HDD50	3400	950	653	350	72	3	0	0	0	3	98	433	838		
	HDD65	6671	1415	1073	801	394	140	14	2	13	146	488	882	1303		
	CDD50	2913	0	0	14	135	370	602	738	641	337	75	1	0		
	CDD65	708	0	0	0	7	42	166	275	188	30	0	0	0		
	CDH74	7112	0	0	4	126	635	1849	2619	1581	286	12	0	0		
	CDH80	2393	0	0	0	28	139	635	1038	491	62	0	0	0		
(14) Wind	WSAvg	5.6	4.9	5.3	6.2	6.6	6.6	6.1	5.7	5.2	5.1	4.9	5.5	5.4		
(15) Precipitation	PrecAvg	10.9	0.1	0.1	0.4	0.7	1.1	1.2	1.9	2.9	1.7	0.8	0.3	0.0		
	PrecMax	19.2	0.4	0.4	2.2	3.3	2.9	3.2	4.3	9.2	3.7	1.9	1.1	0.2		
	PrecMin	5.7	0.0	0.0	0.0	0.0	0.0	0.2	0.6	0.3	0.2	0.1	0.0	0.0		
	PrecStd	3.1	0.1	0.1	0.4	0.7	0.7	0.7	0.9	2.0	0.9	0.5	0.4	0.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.2	57.7	72.4	83.9	87.3	92.1	95.3	91.1	87.5	75.6	63.9	49.8		
		MCWB	33.6	39.4	47.5	53.8	55.5	58.8	63.8	63.8	58.0	52.8	44.4	34.4		
	2%	DB	42.5	50.7	65.2	77.5	83.7	89.1	91.4	88.0	80.5	70.8	57.9	44.0		
		MCWB	30.1	35.6	43.9	50.8	54.7	58.8	63.9	63.8	58.8	51.7	41.5	31.5		
	5%	DB	37.8	45.8	60.2	72.7	80.8	86.5	88.7	85.3	76.8	66.8	53.8	39.8		
		MCWB	27.6	33.1	41.4	48.5	53.9	58.4	63.2	63.9	57.6	49.7	39.6	29.5		
	10%	DB	33.5	42.3	55.6	68.8	77.5	83.8	86.0	82.5	73.6	63.0	49.6	36.0		
		MCWB	25.0	31.3	39.1	47.0	52.5	57.9	62.9	62.9	57.5	48.2	38.0	27.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	33.9	40.6	49.0	56.6	60.4	66.9	71.8	71.2	67.5	57.3	46.8	35.8		
		MCDB	47.2	55.6	67.8	68.6	76.2	77.3	81.3	82.4	72.6	69.1	56.9	46.0		
	2%	WB	30.4	36.9	45.2	53.5	58.3	64.7	69.7	69.3	63.7	54.5	44.0	32.4		
		MCDB	41.5	48.5	61.9	71.2	74.5	77.1	80.9	79.0	72.5	64.7	53.6	43.2		
	5%	WB	28.0	34.3	42.4	51.3	56.7	63.0	68.3	67.9	61.4	52.3	41.4	30.1		
		MCDB	37.4	44.2	57.8	66.8	72.5	75.5	79.7	77.7	70.2	62.6	51.0	38.6		
	10%	WB	25.3	31.8	39.9	49.2	55.0	61.3	66.9	66.6	59.6	50.3	39.0	27.7		
		MCDB	33.1	41.4	53.5	64.1	70.8	74.2	78.3	76.8	68.6	59.7	48.5	35.8		
(35) Mean Daily Temperature Range	MDBR		23.6	24.3	24.0	24.3	23.4	22.3	19.9	19.1	19.5	22.1	22.1	22.0		
	5% DB	MCDBR	29.0	30.2	31.0	30.4	29.5	27.7	24.6	23.0	25.1	26.8	27.7	26.6		
		MCWBR	19.8	19.1	16.6	13.4	11.2	8.2	6.5	6.5	8.8	11.9	15.4	17.6		
	5% WB	MCDBR	27.9	27.5	27.6	24.1	22.7	18.6	17.7	17.1	17.2	22.0	24.1	24.8		
		MCWBR	19.4	18.0	15.6	11.5	9.9	7.0	6.1	6.1	7.6	10.7	14.5	16.8		
(40) Clear-Sky Solar Irradiance	taub		0.351	0.416	0.465	0.500	0.513	0.526	0.505	0.517	0.489	0.448	0.375	0.337		
	taud		2.033	1.883	1.807	1.765	1.803	1.837	1.974	1.916	1.924	1.940	2.037	2.105		
	Ebn at Noon		252	248	249	249	248	245	249	242	240	234	240	248		
	Edh at Noon		40	53	63	69	68	66	57	59	55	49	39	34		
(44) All-Sky Solar Radiation	RadAvg		874	1137	1535	1850	2050	2042	1919	1722	1387	1220	962	809		
	RadStd		82	92	99	129	105	129	98	108	116	104	98	73		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	-1.10	N/A	N/A	N/A	N/A

Nomenclature: See separate page